

Why vit

As in cnn there are some assumptions like nearby pixels more related , translation invariant . but in vit we use self attention that make each patch to interact with each other so we have global relationships as well as local relationships . Which is important for understanding the image where position also matter .

This is particularly useful when the spatial relationship and context between distant regions of an image are important for understanding the overall image.

how image Patchification works

Patchification divides the image into non-overlapping 16×16 patches. It is conceptually similar to using a 16×16 kernel with stride 16, where each kernel position gives us one patch

patch embedding

Patch embedding means converting each flattened image patch into a fixed-dimensional vector using a learnable linear projection. This embedding vector is then given to the Transformer.

positional embedding

In the images position of each patch do matter so in the patch embedding we add position embedding so that we can get both visual embedding and the position of the patch that will help is in further classification . self attention doesnot inherently know the spatial position of the patch .

CLS token

CLS token is a learnable token added at the beginning of the patch sequence. Initially, it has no image-specific information. Through self-attention, it interacts with all the patches and learns an aggregated representation of the whole image, which is then used for classification.

ViT transformer architecture

Images $\rightarrow$ patches $\rightarrow$ flatten patches $\rightarrow$ linear projection $\rightarrow$ embedding vector $\rightarrow$ cls token + positional vector $\rightarrow$ transformer encoder $\rightarrow$ classification with mlp $\rightarrow$ output

CLS vs Patch token

Class token is the learnable token that learns an aggregate representation of the whole image while patch token is just the tokenised form of a particular patch .

Patch token exchange information through the attention layer while the cls learn information through the attention .

Patch token contain visual information of the image .

What dinov2

Combines two objectives dino (knowledge distillation) + masked images modeling

DINOv2 is a self-supervised vision framework that uses teacher–student self-distillation along with masked patch prediction to learn powerful visual representations without requiring labelled images.

What is dinov2 output

DINOv2 gives us rich visual feature embeddings, including patch-level embeddings. These patch features can then be used for downstream tasks like segmentation, classifications . DINOv2 itself does not directly output a segmentation mask; a segmentation method or head is applied to the features to obtain

How dinov2 will be used in this project

Using dinov2 we will do the segmentation of the astronomical images. Dinov2 will be feature extractor it will give us patch embeddings which will be converted into the segmentation mask.